CUSTOMER COHORT & CHURN ANALYTICS

Internship Task 3 | IBM Telco Customer Churn Dataset

Project Report

This report analyzes customer churn patterns using the supplied telecom customer snapshot. It calculates the overall churn rate, compares churn across tenure bands, contract types, payment methods, internet service and support options, and translates the findings into practical retention recommendations.

7,043	1,869	26.54%	73.46%
Total customer records	Churned customers	Overall churn rate	Observed retention rate

Executive Summary

The dataset contains 7,043 customer records and 22 columns. 1,869 records are labeled as churned, giving an observed churn rate of 26.54%. The remaining 5,174 records are labeled as not churned, equivalent to 73.46% observed retention in this snapshot. The analysis below highlights which customer segments have higher churn rates; these associations should guide investigation rather than be treated as proof of causation.

Business objective

Help a telecom/SaaS business understand which customer groups show higher churn and identify retention actions that can be tested and measured.

Dataset and methodology

The supplied CSV was analyzed with Python and pandas. Churn rate is calculated as churned customers divided by customers in the relevant group, multiplied by 100. Tenure cohorts are created from current tenure months. Duplicate rows and missing values should be reviewed before production use; TotalCharges is converted to numeric for quality checking.

1. Overall Churn

The overall churn rate is the baseline against which segment rates can be compared. A higher segment rate identifies a group for further investigation, but customer counts and business context should also be considered.

Metric	Value
Customer records	7,043
Churned (Churn = Yes)	1,869
Not churned (Churn = No)	5,174
Observed churn rate	26.54%
Observed retention rate	73.46%

2. Tenure-Based Cohort Analysis

Customers are grouped by their current tenure: 0–12, 13–24, 25–48, 49–60, and 61+ months. This helps identify tenure bands with comparatively high observed churn.

Current tenure cohort	Customers	Churned	Churn rate	Retention rate
0–12 months	2,186	1,037	47.44%	52.56%
13–24 months	1,024	294	28.71%	71.29%
25–48 months	1,594	325	20.39%	79.61%
49–60 months	832	120	14.42%	85.58%
61+ months	1,407	93	6.61%	93.39%

Interpretation: Use this table to prioritize early-tenure support if the shortest-tenure cohort has the highest churn. This is a tenure-segment comparison, not a true acquisition-month cohort retention curve.

3. Churn by Contract Type

Contract length can be associated with differences in customer commitment and cancellation flexibility. Compare churn rate and customer count for each contract group.

Contract	Customers	Churned	Churn rate
Month-to-month	3,875	1,655	42.71%
One year	1,473	166	11.27%
Two year	1,695	48	2.83%

Interpretation: If month-to-month contracts have the highest churn, test clearer value communication, proactive service check-ins, and optional incentives for longer commitments. Do not assume that forcing longer contracts will improve customer experience.

4. Churn by Payment Method

Payment method may correlate with billing preferences, payment friction, or differences between customer groups. Review the churn rate alongside the number of customers in each category.

Payment method	Customers	Churned	Churn rate
Electronic check	2,365	1,071	45.29%
Mailed check	1,612	308	19.11%
Bank transfer (automatic)	1,544	258	16.71%
Credit card (automatic)	1,522	232	15.24%

Recommended follow-up: make payment options easy to understand, review billing-related complaints, and test reminders or payment-method guidance where appropriate. Segment-level correlation does not establish payment method as the cause of churn.

5. Churn by Internet Service and Support

Internet service and support options can reveal customer groups that may need closer service-quality investigation. The tables below show the observed churn rate for each category.

Internet service	Customers	Churned	Churn rate
Fiber optic	3,096	1,297	41.89%
DSL	2,421	459	18.96%
No	1,526	113	7.40%

Tech support	Customers	Churned	Churn rate
No	3,473	1,446	41.64%
Yes	2,044	310	15.17%
No internet service	1,526	113	7.40%

Recommended follow-up: compare complaints, outage records, installation issues, and support interactions for higher-churn service groups before choosing an intervention.

6. Monthly Charges and Risk Segments

Monthly charges should be analyzed in bands or alongside service type and contract type. A relationship between higher charges and churn may reflect service mix or customer needs rather than price alone. The accompanying notebook includes a charge-band analysis and a segment summary table.

7. Recommendations

Prioritize the early customer journey

Offer onboarding check-ins, setup guidance, and early issue resolution to customers in the tenure bands with the highest observed churn.

Investigate flexible contracts

If month-to-month customers show higher churn, test transparent plan comparisons and optional longer-term incentives; monitor both churn and customer satisfaction.

Review billing friction

Examine payment-related complaints and test clear billing explanations, reminders, and accessible payment options.

Improve service and support experience

Use support contacts, service incidents, and satisfaction data to determine whether high-churn service groups face specific issues.

Measure interventions

Run small controlled tests where possible. Track churn, retention, complaints, and revenue impact before scaling a retention action.

Avoid causal claims from a snapshot

Treat the observed relationships as hypotheses to investigate. Validate with additional time-based and operational data.

8. Dashboard Design

Suggested Power BI or Python dashboard components:

Visual	Purpose
KPI cards	Total customers, churned customers, churn rate, retention rate
Bar chart	Churn rate by contract type
Bar chart	Churn rate by payment method
Tenure cohort chart	Churn and retention rate by current tenure band
Service/support heatmap	Compare churn by internet service and tech support
Monthly charge band chart	Compare churn rates across charge ranges
Filters/slicers	Contract, tenure cohort, internet service, payment method

9. Limitations

This file is a customer-level snapshot and does not include customer acquisition dates or month-by-month activity history. Therefore, it cannot independently support a true calendar-month cohort retention matrix or survival curve. The report calculates observed churn rates; it does not train a predictive churn model or produce individual churn probabilities. Missing or blank TotalCharges values should be reviewed, and any further analysis should document the handling of those rows.

10. Conclusion

The dataset's observed overall churn rate is 26.54%. Segment analysis provides a practical starting point for understanding where churn is concentrated. The next step is to validate potential drivers with customer feedback, support history, and time-based data, then test retention actions and measure their effect.

Prepared for: Internship Task 3 — Customer Cohort & Churn Analytics

Tools: Python, pandas, matplotlib/seaborn; dashboard can be implemented in Power BI Desktop or Python.